You might think there's a loophole: What if Alice doesn't just measure her qubit and find out a random answer, but rather forces her answer to be spin-up? Then Bob would also get spin-up. That would seem like information had been transmitted instantaneously.

The problem is that there's no straightforward way to start with a quantum system that is in a superposition and measure it in such a way that we can force a particular answer. If Alice simply measures her spin, she'll get up or down with equal probabilities, no ifs, ands, or buts. What Alice can do is to manipulate her spin before she measures it, forcing it to be 100 percent spin-up rather than in a superposition. For example, she can shoot a photon at her electron, with just the right properties that the photon leaves the electron alone if the electron was spin-up, and flips the electron to spin-up if it was spin-down. Now Alice's original electron will definitely be measured to be spin-up. But that electron is also no longer entangled with Bob's electron. Rather, the entanglement has been transferred to the photon, which is in a superposition of "left Alice's electron alone" and "bumped into Alice's electron." Bob's electron is completely unaffected, and he's going to get spin-up or spin-down with fifty-fifty probability, so no information has been transmitted.

This is a general feature of quantum entanglement: the *no-signaling theorem*, according to which an entangled pair of particles cannot actually be used to transmit information between two parties faster than light. So quantum mechanics seems to be exploiting a subtle loophole, violating the spirit of relativity (nothing travels faster than the speed of light) while obeying the letter of the law (actual physical particles, and whatever useful information they might convey, cannot travel faster than the speed of light).

The so-called *EPR paradox* (which isn't a paradox at all, just a feature of quantum mechanics) goes beyond simple worries about spooky action at a distance. Einstein aimed to show not only that quantum mechanics was spooky but that it couldn't possibly be a complete theory—that there had to be some underlying comprehensive model for which quantum mechanics was simply a useful approximation.